AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 30 · STUDENT LESSON

Choose Inputs and Outputs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Weeks 2 and 16: specify an information path and an uncertainty response.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Turn the project outcome into precise input and output requirements. Try the recall before opening notes.

Your learning target

I can turn the project outcome into precise **input** and **output requirement**s.

I can apply a simple manual **baseline** to valid requests and record its results.

I can plan a fair comparison using the same catalogue, requests and judging rules.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

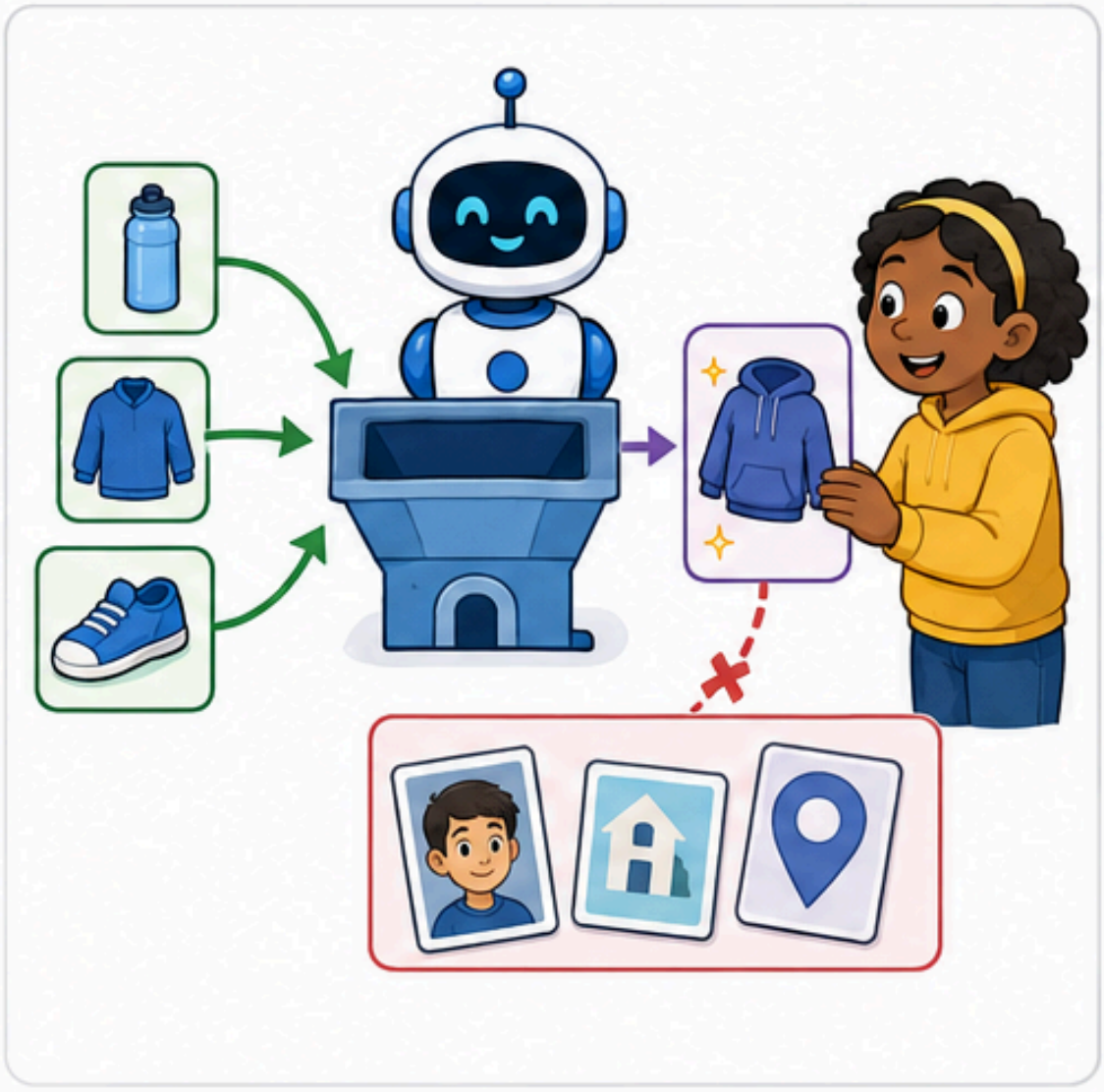
Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Verify Important **Output**s. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A useful design makes required information and promised result explicit.



Your target

LEARNING TARGET Design a paper/plastic sorter that receives an object description. Include a mixed paper-and-plastic cup. Specify **input**s, **output**s and the uncertain-case action. Is the owner's name useful?

Learner name:

Readiness for the connected explanation

A recommender always reports no match. Why is a count of zero in**compatible** suggestions insufficient to show that it helps users?

This week’s end goal

By the end, I can specify valid **input**s, exact **output**s and a manual **baseline**.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Choose Inputs and Outputs case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY A sorter can use item type and location found to suggest a bin without collecting a student's name or face.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Last week we identified a difficulty: some fictional activity choices did not fit the available time or room conditions. Now define the task precisely enough that two reviewers can agree whether an **output** is correct.

A **requirement** such as “suggest a good activity” is too vague. State which **input**s are accepted, what counts as **compatible** and what to do when no option fits.

For this paper project, a valid request contains an available time from 0 to 30 whole minutes and a quiet-room **requirement** written Yes or No. Yes means the activity must be marked quiet.

No means either quiet or non-quiet activities are permitted; it does not mean noise is required. A missing or invalid value needs clarification, not a guessed value or a no-match claim.

A **compatible** activity takes no more than the available minutes and meets the quiet-room condition. If several activities fit, suggest the first in catalogue order. If none fits a valid request, report no match.

Give a brief reason. This is a suggestion only: it does not book resources, tell anyone to begin or guarantee enjoyment.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Choose Inputs and Outputs. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



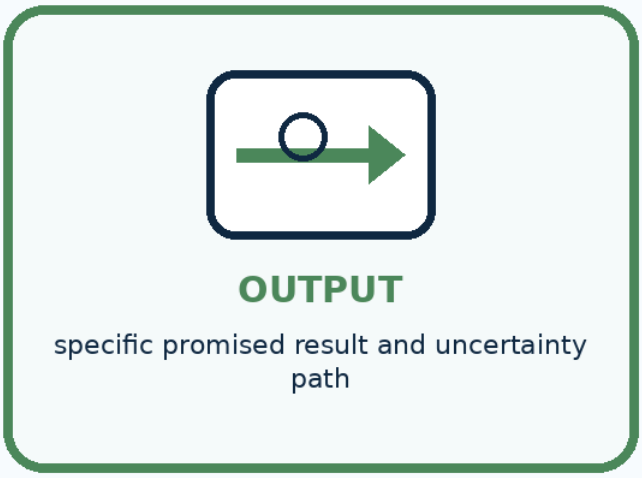
INPUT

minimal required information



PROCESS

candidate decision process



INPUT minimal required information

PROCESS candidate decision **process**

OUTPUT specific promised result and uncertainty path

Key idea

CORE IDEA A useful design makes required information and promised result explicit.

How the mechanism works

UNDER THE HOOD Interface contracts define **input**s, **output**s, types, allowed values, **missing-data behavior**, and guarantees.

Good design avoids collecting data that cannot improve the promised result and avoids promising outcomes the **process** cannot support.

Draw a fresh example with all three stages:

A closer explanation

Catalogue order and activity	Minutes required	Quiet activity
B1 Bookmark drawing	10	Yes
B2 Story circle	8	No
B3 Pattern puzzle	12	Yes
B4 Tower challenge	15	No

These four rows are the complete approved catalogue for the paper model. Treat their times as fixed task values, not measured guarantees about real pupils. All required materials are assumed available.

The catalogue order B1, B2, B3, B4 is the **tie rule**; it is not a claim that B1 is everyone's favourite.

For 10 minutes with quiet required, B1 fits because 10 is no more than 10 and B1 is quiet. B2 fails the quiet **requirement**. B3 and B4 take too long. For 8 minutes with quiet not required, B2 fits; B1 takes too long.

A request for 7 minutes has no **compatible** activity in this catalogue, even when quiet is not required.

Check each condition. A fast activity can still fail the quiet **requirement**. A quiet activity can still take too long. Exact equality is allowed: an activity listed as 10 minutes fits 10 available minutes.

A blank time is different from a valid time of zero.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Choose Inputs and Outputs. Identify the detail that changes the conclusion.

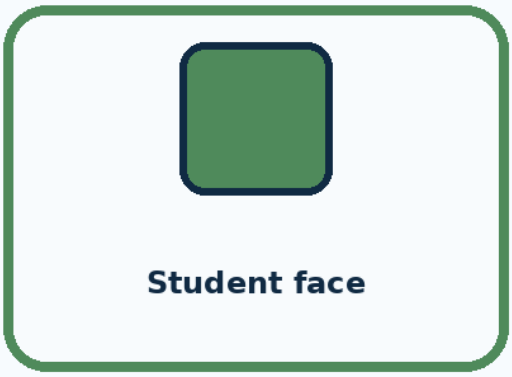
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Item type Clue: directly supports bin choice. Meaning: required candidate.

The case to investigate

CASE 2 — Location found Clue: may disambiguate. Meaning: optional candidate.

The case to investigate

CASE 3 — Student face Clue: not needed. Meaning: remove.

The case to investigate

CASE 4 — Guaranteed owner Clue: unsupported promise. Meaning: revise **output**.

Keep this boundary clear

BOUNDARY Every collected field creates cost and risk; every promised **output** needs **evidence**.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Change request 6 to 5.

Predicted result

Prediction: The **baseline** returns B1.

Why the result follows

Reason: Its capacity equals the request, and at least includes equality.

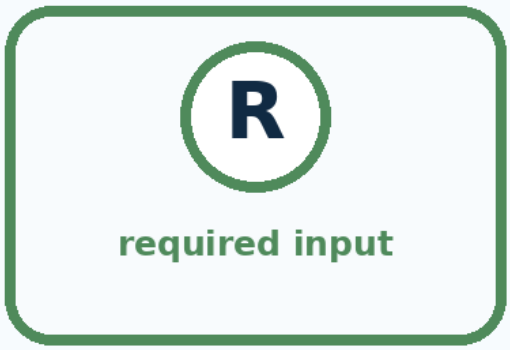
Words that unlock the idea

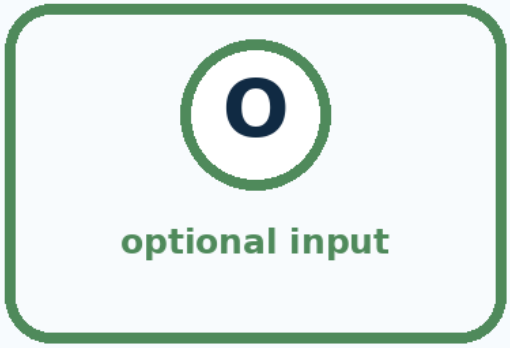
> Use the vocabulary of Choose Inputs and Outputs to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
required input	information necessary for the process	
optional input	information that may help but is not required	
output contract	specific result the system promises to produce	
missing-data behavior	what happens when an input is absent	

Misconception detective

Find the unsupported claim

CLAIM TO REPAIR More **input**s make every design more accurate.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Requirement	A precise condition that the solution must satisfy
Baseline	A defined starting method used for comparison
Compatible	Meeting all the stated conditions of the request
Tie rule	A stated way to choose when several options qualify
Acceptance check	A check of whether a result satisfies a requirement

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STUDENT LESSON · 6 OF 18

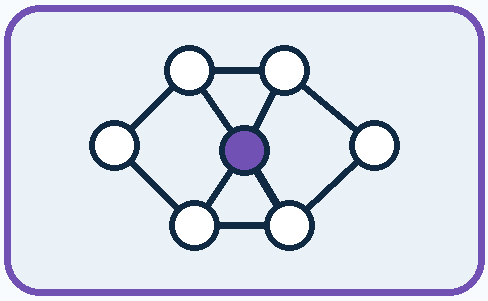
Follow a worked example

> Follow the worked trace for Choose Inputs and Outputs, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE A sorter can use item type and location found to suggest a bin without collecting a student's name or face.

Step 1

STEP 1 — OBSERVE minimal required information

Step 2

STEP 2 — TRACE candidate decision **process**

Step 3

STEP 3 — CONCLUDE specific promised result and uncertainty path

Model answer

MODEL REASONING Prefer the design that meets the measurable need with fewer, clearer **input**s and an honest **output**. State how missing values and ambiguous cases are handled before building.

A claim to avoid

DO NOT OVERCLAIM Every collected field creates cost and risk; every promised **output** needs **evidence**.

What new **evidence** could change this answer?

A closer explanation

The manual checklist baseline

STAGE	MANUAL ACTION	RECORDED RESULT
Validate the request	Check minutes and Yes or No	Clarify invalid input
Inspect catalogue in order	Check time and quiet for each	Mark compatible rows
Choose or explain absence	First compatible row or none	Suggestion or no match
Keep the evidence	Record input, result, reason	Comparable baseline record

The baseline is a complete manual method. A prototype must be compared with this defined procedure, not an invented weak opponent.

Worked explanation · Choose Inputs and Outputs

Write the request at the top of a sheet. If it is invalid, record what needs clarification and stop. Otherwise mark each catalogue row as **compatible** or in**compatible** using both conditions.

Circle the first **compatible** row in order, or write no match if every row fails. Record the **output** and its reason. This checklist is our **baseline**.

Worked explanation · Choose Inputs and Outputs

The **baseline** can be correct on all cases when applied carefully. A prototype that matches it has not demonstrated better compatibility.

It may still be easier to use or faster to trace, but those are separate claims that need their own measures. Do not make the **baseline** skip a condition just to make the prototype look better.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Choose Inputs and Outputs

A fictional classroom chooses a storage bin for model parts.

The facts we will use

Supplied facts

Catalogue order: B1 capacity 5, B2 capacity 9. Valid request is a whole part count 1–12.

Supplied facts

Return first bin with capacity at least the count; none if neither fits; clarify invalid **input**.

AI ACADEMY / AI-1/30

Choose Inputs and Outputs

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.

1. Validate request
Count 6 is whole and within 1-12.

↓

2. Check first bin
B1 capacity 5 is below 6; reject it.

↓

3. Check second bin
B2 capacity 9 is at least 6; accept it.

↓

4. Baseline output
Report B2 and stop; save the comparison evidence.

CHANGE ONE THING
Change request 6 to 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Validate request	Count 6 is whole and within 1–12.
2. Check first bin	B1 capacity 5 is below 6; reject it.
3. Check second bin	B2 capacity 9 is at least 6; accept it.
4. Baseline output	Report B2 and stop; save the comparison evidence.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The **baseline** is an ordered catalogue scan, not a preference ranking. It makes expectations testable before a prototype is built.

Count 10 is valid but has no matching bin; that differs from a blank count, which fails the **input** contract. Both situations need explicit **output** behavior.

What this example does not establish

Keep the stated limit

Bin capacity is the only supplied **fitting** criterion; weight or shape suitability cannot be inferred.

Keep the stated limit

Read the labelled picture: Trace requests 10 and blank. Why should they not receive the same explanation?

Keep the stated limit

Change one thing: Change request 6 to 5. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Choose Inputs and Outputs claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Item type	directly supports bin choice	required candidate
Location found	may disambiguate	optional candidate
Student face	not needed	remove
Guaranteed owner	unsupported promise	revise output

Repair two tempting claims

Challenge claim A

CLAIM A More **input**s make every design more accurate.

Repair:

Challenge claim B

CLAIM B An **output** label can promise more than the **process** actually knows.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

Your project record now needs the problem brief, this catalogue, the **input** rules, compatibility rule, **tie rule** and clarification behaviour.

Add a **baseline** sheet with one row per request: supplied **input**, actual **output**, reason and whether the **requirement**s were met. Record the method and catalogue version so a later reader can repeat the comparison.

Check the claim against the facts

For this week, complete the **baseline** before building a new **process**. Next week you will design the full decision path, including invalid **input** and no-match cases. Keep compatibility claims within the paper model.

Real materials, changing room conditions and pupils' preferences would require **more information** and a revised task.

Two methods both produce the required outputs on the same five valid requests. One team claims its prototype is more accurate because it uses an app. Is that conclusion supported? Name a separate benefit the team could investigate and what it would need to record

STUDENT LESSON · 8 OF 18

Practise together

Practise Choose Inputs and Outputs with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A useful design makes required information and promised result explicit.

Use a fresh example to explain **required input**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: An **output** label can promise more than the **process** actually knows.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Both methods must receive the same request values and the same catalogue version, and both must be judged by the same **requirement**s. Record actual results, including errors, no-match **output**s and clarification requests.

A valid no-match result can pass; a no-match result that overlooks a **fitting** activity cannot. A guessed answer to invalid **input** fails the clarification **requirement**.

> **Apply the manual baseline to 12 minutes with quiet required and to 8 minutes with quiet required. Mark every catalogue row for each request, choose the required output and explain the tie rule. State what to do if the available-time field is blank instead**

The cases used while designing are development cases. Keep the teacher's reserved final cases unopened until the **process** is fixed and ready for evaluation.

Do not change the **requirement**s after seeing a poor result simply to count that result as a success. If a real need changes, record the new **requirement** and rerun the relevant checks.

Practise with less help: Run the baseline on three requests

Use workbook W2 for the connected example "**Requirement**s and a simple **baseline**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Requirement**s and a simple **baseline**. Proposed explanation: "Quiet required No excludes quiet activities". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Trace requests 10 and blank. Why should they not receive the same explanation?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

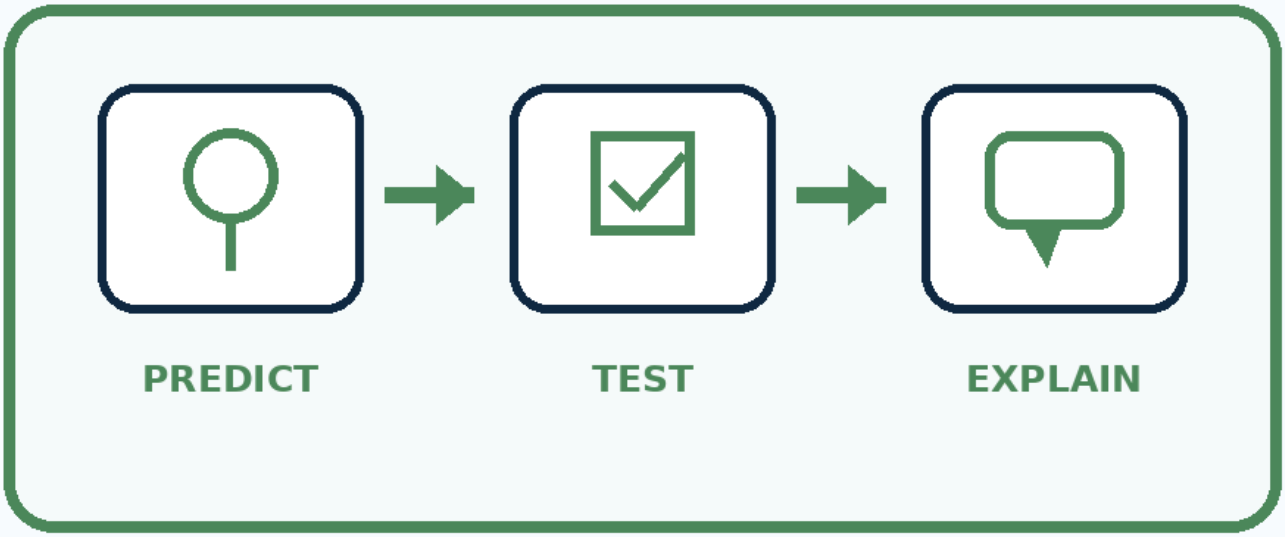
Run the investigation

> Investigate Choose Inputs and Outputs. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Two-Design Input Audit

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Sketch two designs and remove an unnecessary **input**.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

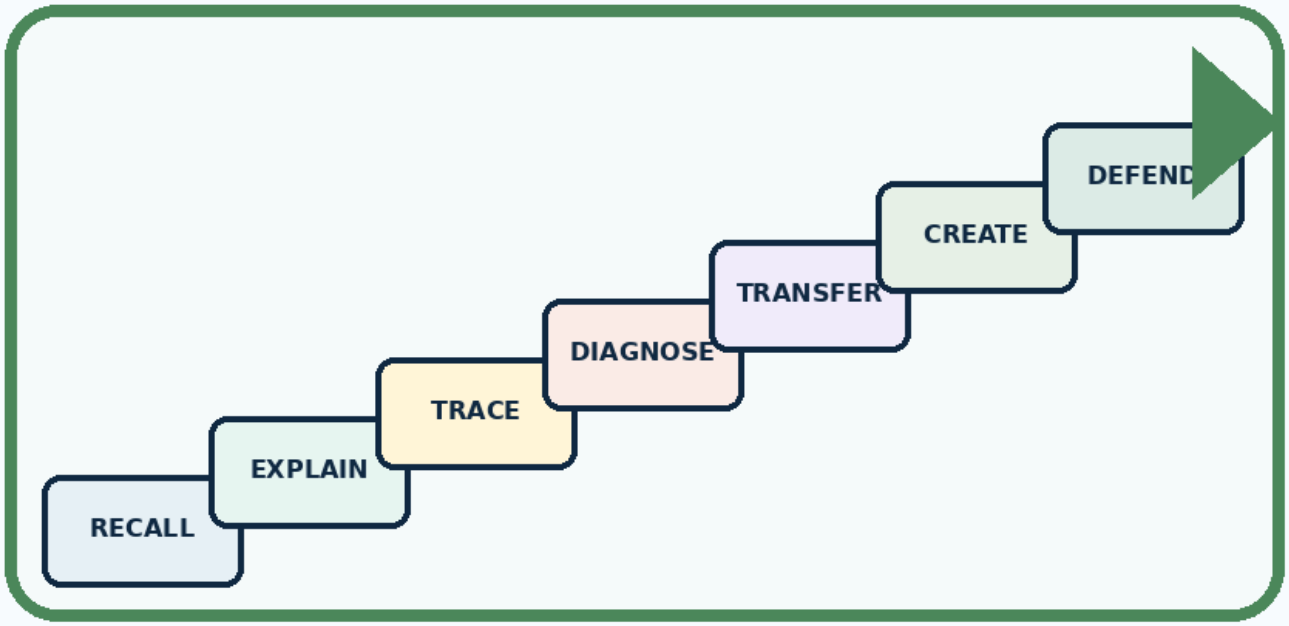
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Choose Inputs and Outputs. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **required input**.

EXPLAIN A useful design makes required information and promised result explicit.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: More **input**s make every design more accurate.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

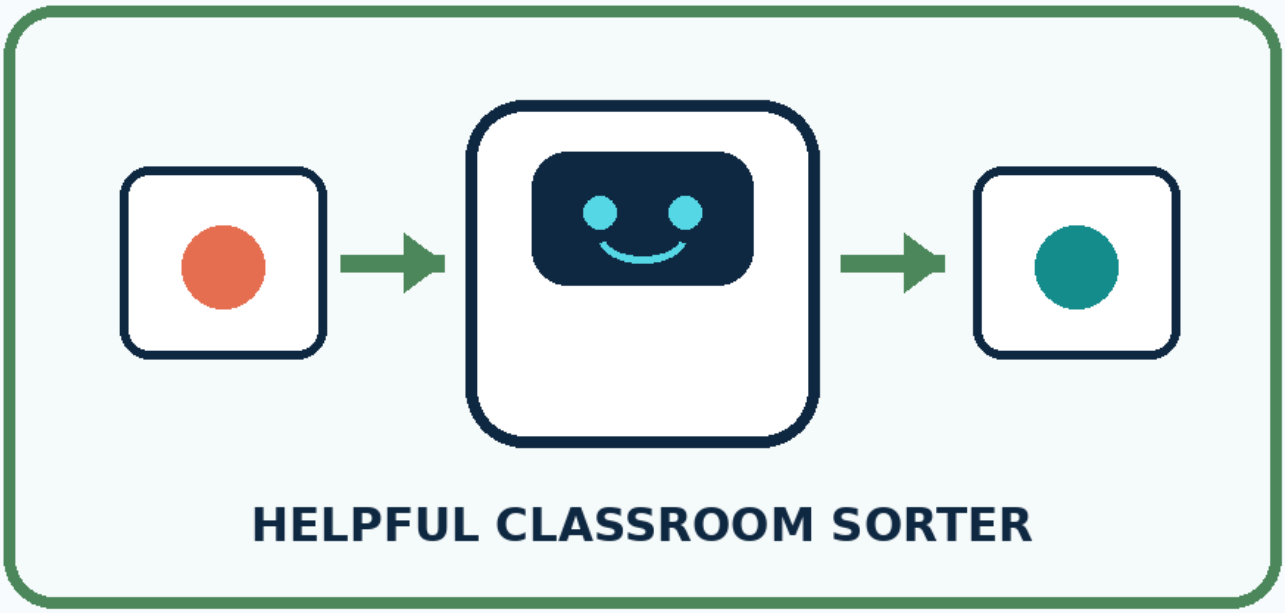
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Freeze the sorter's v1 **input** list, **output contract**, and not-sure behavior.

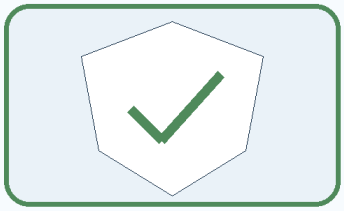
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN Every collected field creates cost and risk; every promised **output** needs **evidence**.

My project decision:

Evidence for your project

Document sorter schema, catalogue order, acceptance rule and all **output** types; use a manual trace as the comparison **baseline**.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Sketch two designs, compare their **input** and **output contract**s, and remove one unnecessary **input** with a reason.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK Prefer the design that meets the measurable need with fewer, clearer **input**s and an honest **output**. State how missing values and ambiguous cases are handled before building.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Design the Decision **Process**.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Choose Inputs and Outputs case for supported practice. Repeated examples do not become fresh assessments.

Week 30 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A sorter suggests paper or plastic trays using material **observation**s.

The question

Name an **input**, **output** and unnecessary **input**.

Worked explanation

Input: material or a suitable image. **Output**: tray suggestion. Home address is unnecessary. Uncertain material needs review.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Design a sorter with only relevant inputs

Evidence or stage	Value or result
Input	Material or a suitable image
Decision	Paper, plastic or uncertain
Output	Tray suggestion
Review	Ask a person when material is unclear

A sorter suggests paper or plastic trays using material **observation**s.

Name an **input**, **output** and unnecessary **input**.

Input: material or a suitable image. **Output**: tray suggestion. Home address is unnecessary. Uncertain material needs review.

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Build a baseline for a different catalogue

A paper room recommender uses R1 first, capacity 6, quiet Yes; then R2, capacity 10, quiet No. Valid requests give a whole group size from 1 to 12 and Quiet required Yes or No.

A room fits when capacity is at least group size and the quiet condition is met; choose the first **fitting** room. Record **output**s and reasons for 6 people Yes, 8 people No and 8 people Yes.

Explain how a manual **baseline** can be compared fairly with a prototype. No real booking is allowed.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 30 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A paper sorter must suggest pencil tray or book shelf. Choose one needed **input** and state the **output**. What should happen if the picture is blocked?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Freeze the sorter's v1 **input** list, **output contract**, and not-sure behavior.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

> Design a paper/plastic sorter that receives an object description. Include a mixed paper-and-plastic cup. Specify **input**s, **output**s and the uncertain-case action. Is the owner's name useful?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Choose Inputs and Outputs, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional classroom chooses a storage bin for model parts.

By the end, I can	How I will show it
Turn the project outcome into precise input and output requirements	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Apply a simple manual baseline to valid requests and record its results	Explain the deciding step, show a trace or test, and state one limit.
Plan a fair comparison using the same catalogue, requests and judging rules	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Choose Inputs and Outputs without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Turn the project outcome into precise **input** and **output requirement**s

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional classroom chooses a storage bin for model parts.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Report B2 and stop; save the comparison **evidence**.

1. **Baseline output**

2. Check first bin

3. Validate request

4. Check second bin

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Baseline output** Report B2 and stop; save the comparison **evidence**. The **baseline** is an ordered catalogue scan, not a preference ranking.

It makes expectations testable before a prototype is built. Count 10 is valid but has no matching bin; that differs from a blank count, which fails the **input** contract. Both situations need explicit **output** behavior.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change request 6 to 5. Which conclusion is supported by the supplied facts?

1. The **baseline** returns B1.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The **baseline** returns B1. Its capacity equals the request, and at least includes equality.

Repair a missing building block

Review Find a Real Problem (ai-1/29). Goal: Describe a specific user difficulty using supplied **evidence** without inventing its cause.

Short practice: Write a one-sentence project goal and say what **evidence** would count as a failure.. Return to this week and explain what you changed.

Review Verify Important **Output**s (ai-1/27). Goal: Separate an answer into claims and connect each claim to relevant **evidence**.

Short practice: Correct the draft and identify which part needs another source rather than a calculation.. Return to this week and explain what you changed.

Review Privacy by Choosing Less (ai-1/24). Goal: Justify which information a stated task needs and remove unnecessary fields.

Short practice: For the original task, does adding an anonymous-looking pupil code improve the repair count?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

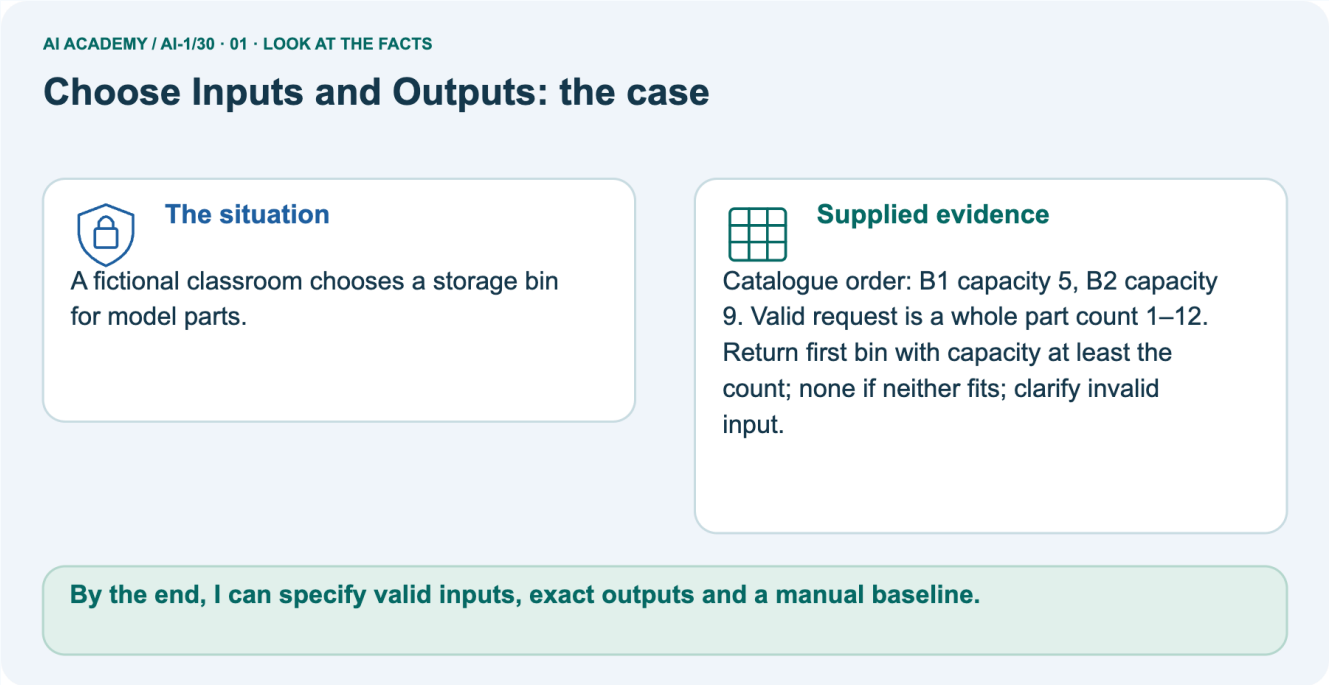
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Choose Inputs and Outputs graphic using its labels, deciding evidence and relevant limitation.

01. Choose Inputs and Outputs: the case



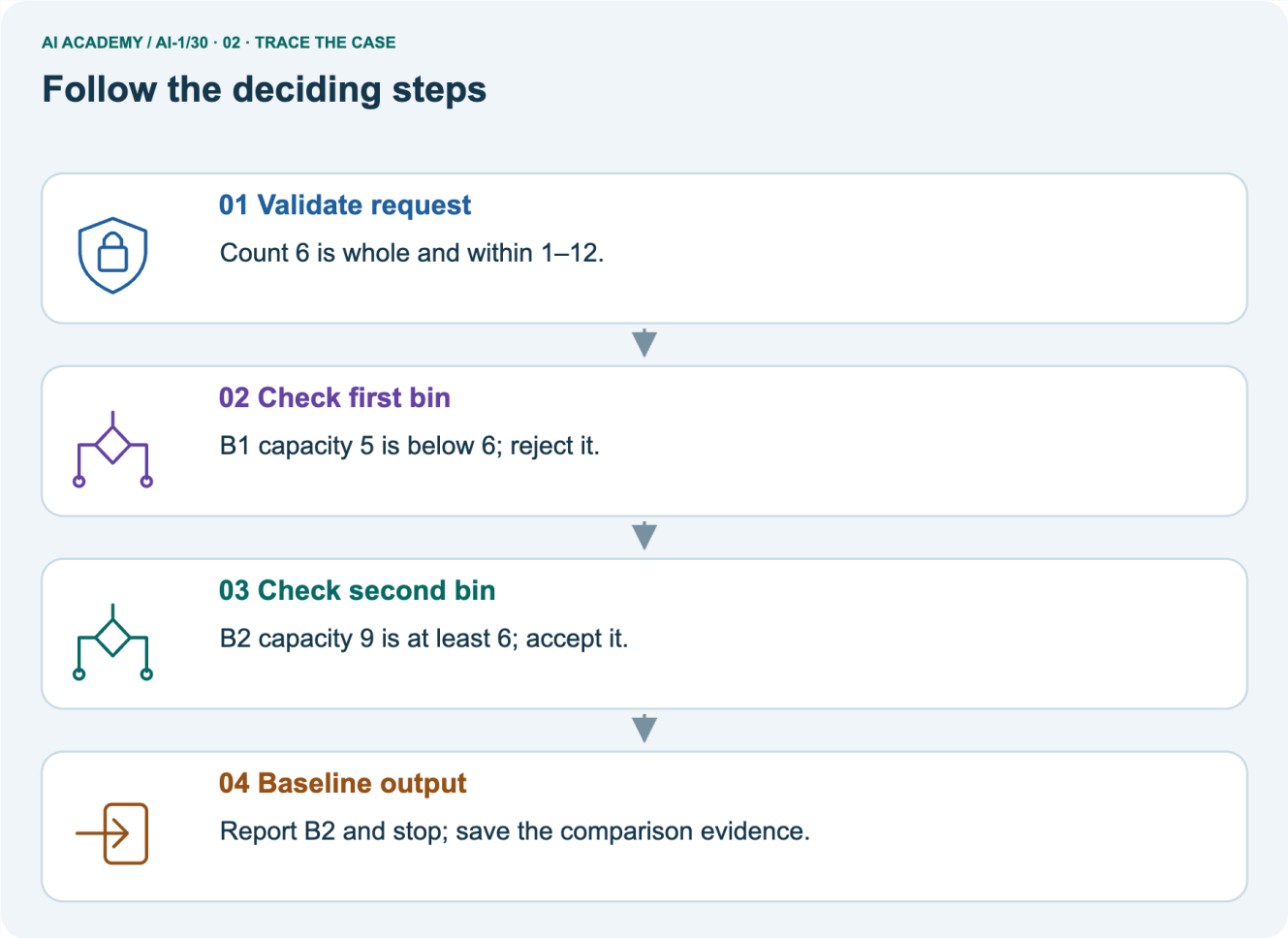
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional classroom chooses a storage bin for model parts. Supplied facts: Catalogue order: B1 capacity 5, B2 capacity 9. Valid request is a whole part count 1–12.

Return first bin with capacity at least the count; none if neither fits; clarify invalid **input**.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate request: Count 6 is whole and within 1–12. → Check first bin: B1 capacity 5 is below 6; reject it. → Check second bin: B2 capacity 9 is at least 6; accept it. → **Baseline output**: Report B2 and stop; save the comparison **evidence**.

> Point to each numbered stage and explain which recorded **input** it uses.

03. The manual checklist baseline

AI ACADEMY / AI-1/30 · 03 · INSPECT THE LESSON MODEL

The manual checklist baseline

Stage	Manual action	Recorded result
Validate the request	Check minutes and Yes or No	Clarify invalid input
Inspect catalogue in order	Check time and quiet for each	Mark compatible rows
Choose or explain absence	First compatible row or none	Suggestion or no match
Keep the evidence	Record input, result, reason	Comparable baseline record

The baseline is a complete manual method. A prototype must be compared with this defined procedure, not an invented weak opponent.

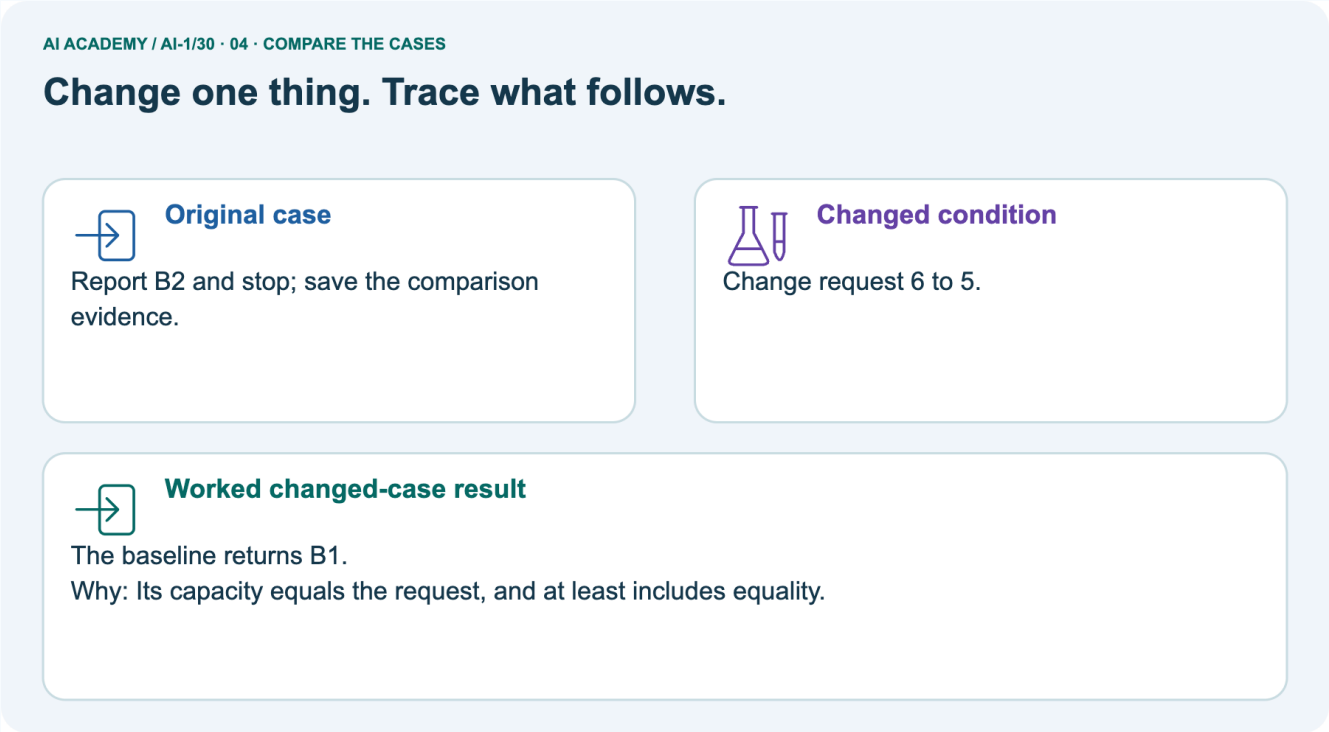
Follow the labelled picture with the matching explanation. Open the vector diagram

The manual checklist **baseline**. Validate the request | Check minutes and Yes or No | Clarify invalid **input**. Inspect catalogue in order | Check time and quiet for each | Mark **compatible** rows.

Choose or explain absence | First **compatible** row or none | Suggestion or no match. Keep the **evidence** | Record **input**, result, reason | Comparable **baseline** record

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.

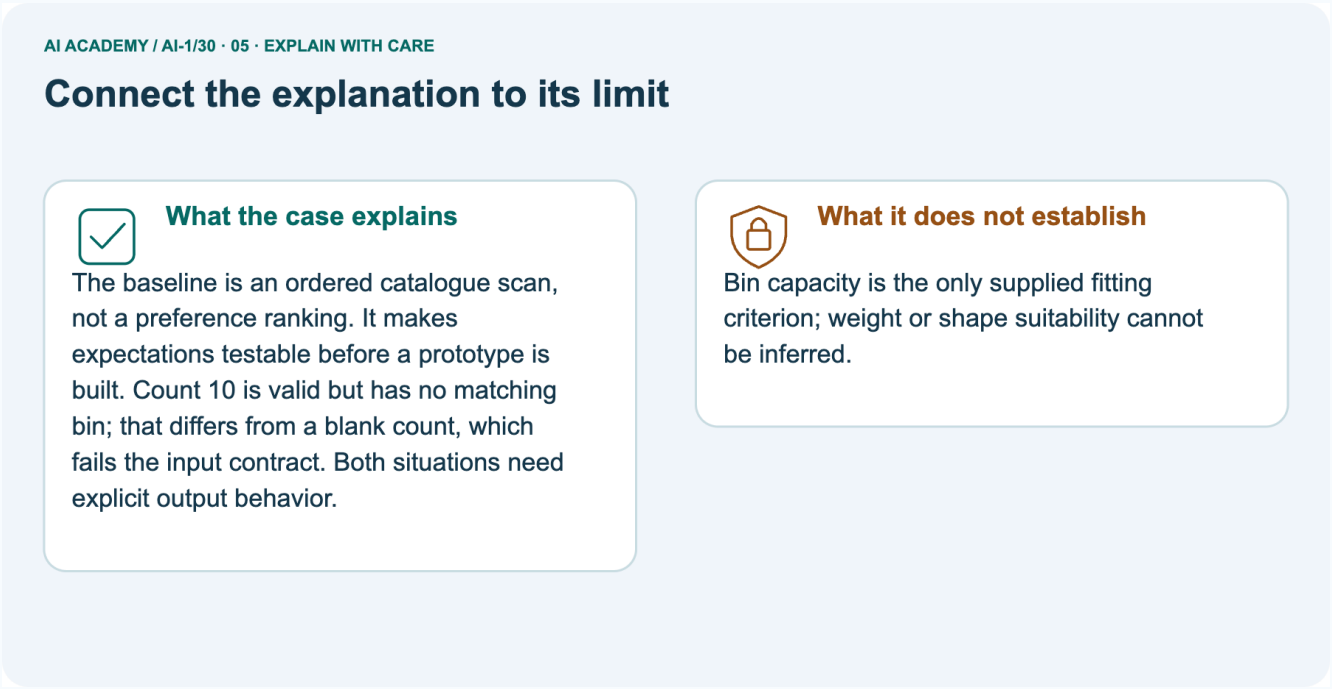


Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Report B2 and stop; save the comparison **evidence**. Changed condition: Change request 6 to 5. Expected result: The **baseline** returns B1. Reason: Its capacity equals the request, and at least includes equality.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

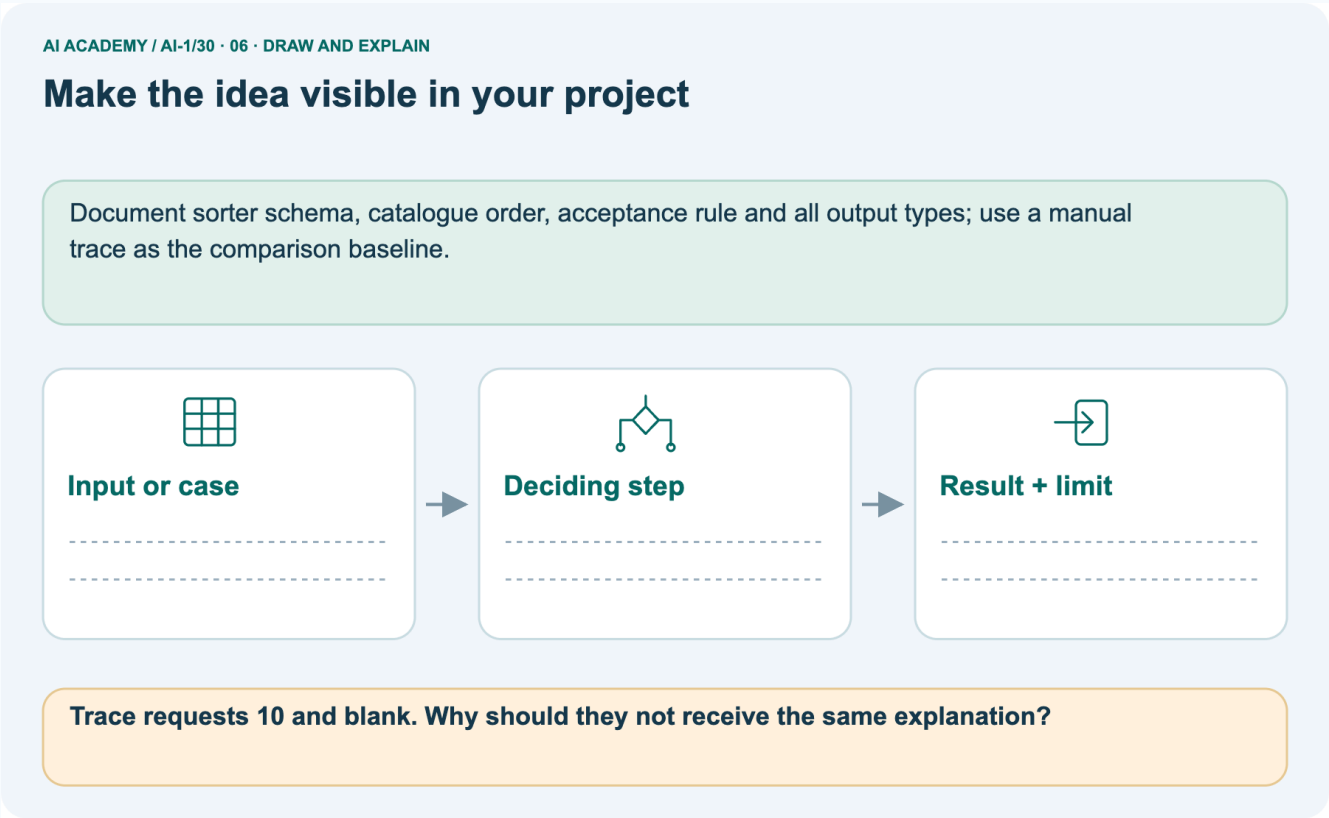
The **baseline** is an ordered catalogue scan, not a preference ranking. It makes expectations testable before a prototype is built.

Count 10 is valid but has no matching bin; that differs from a blank count, which fails the **input** contract. Both situations need explicit **output** behavior.

Limit: Bin capacity is the only supplied **fitting** criterion; weight or shape suitability cannot be inferred.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document sorter schema, catalogue order, acceptance rule and all **output** types; use a manual trace as the comparison **baseline**. Draw your **input**, deciding step, result and limit. Supported question: Trace requests 10 and blank.

Why should they not receive the same explanation?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.